Department of Chemical & Biomolecular Engineering
North Carolina State University

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Discrete Variables & Probability Distributions†

Discrete Variables

Discrete variables are variables that take on integer values, examples include number of packages of a defective product in a production batch, numbers of cars failing an emission test, number of persons afflicted by a certain disease, and number of heads (or tails) turning up in a specified number of coin tosses. Every discrete variable x has a probability distribution, *P(x)*, which is given by the relative frequency distribution *f*, $P(x) = f(x) \equiv \frac{n(x)}{N}$ as $N \to \infty$ (1)

In Eqn (1), n and N denote the frequency of occurrence and number of samples taken, respectively. For a small sample population, $N < 10$, the relative frequency *f(x)* gives a crude estimate of the probability distribution *P(x)* of the discrete variable *x*, but as N becomes larger, *f(x)* gives a progressively better estimate of *P(x)*. In the limit of $N \to \infty$ (or with $N > 500\text{-}1000$ in practice), *f(x)* gives an exact estimate of *P(x)*.

The probability distribution of x is used to determine the expectation (or average) values of the discrete variable x and any function *y(x)* of x, viz., $E(x) \equiv < x > = \sum_{i=1}^{N} P_i(x_i)x_i$ (2)

$$E[y(x)] \equiv < y(x) > = \sum_{i=1}^{N} P_i(x_i)y_i(x_i) \qquad (3)$$

For the special case of $y(x) = [x - E(x)]^2$, the expectation value of $[x - E(x)]^2$ is defined as the variance σ^2 of x and the square root of σ^2 is defined as the standard deviation σ of x, i.e.,

$$\text{Variance of } x \equiv \sigma^2 \equiv E\{[x - E(x)]^2\} = \sum_{i=1}^{N} P_i(x_i)[x_i - E(x)]^2 \qquad (4)$$

$$\text{Standard deviation of } x \equiv \sigma \equiv \sqrt{E\{[x - E(x)]^2\}} = \sqrt{\sum_{i=1}^{N} P_i(x_i)[x_i - E(x)]^2} \qquad (5)$$

A probability distribution function for a discrete variable has the following properties:

(i) $0 \le P(x) \le 1$, and (ii) $\sum_{i=1}^{N} P_i(x_i) = 1.0$ (the probability distribution is said to be normalized) .

Some Known Probability Distribution Functions of Discrete Variables

(A) Binomial Probability Distribution

A binomial trial is an experimental run that has only two outcomes, success and failure (or affirmative and negative), with a constant probability of success p (and, hence, a constant probability of failure, $1-p$). A binomial experiment is one with n repeat trials, each of which is independent and each has a constant probability of success (or failure). The probability distribution of a binomial variable X in n trials is given by $P(x) = {}^{n}C_x p^x (1-p)^{(n-x)}$ (6)

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where $^{n}C_{x} \equiv \dfrac{n!}{x!(n-x)!}$ (7)

$P(x)$ thus gives the probability of x successes in n binomial trials.

Example 1: A fair coin is tossed three times. Let x denotes the number of heads turning up in the three tosses. Find the probability distribution for all possible values of x.

Solution

For a fair coin, the head and tail have equal probabilities of turning face up. Therefore, the probability of head turning up on a toss = 0.5.

x, # Heads Up	**Possible Toss Outcome(s)**	**Probability**
0	TTT	$P(x=0) = (1-0.5)^3 = 0.125$
1	HTT, THT, TTH	$P(x=1) = 3*0.5^1(1-0.5)^2 = 0.375$
2	HHT, HTH, THH	$P(x=2) = 3*0.5^2(1-0.5)^1 = 0.375$
3	HHH	$P(x=3) = 0.5^3 = 0.125$

Note: $\displaystyle\sum_{i}^{\text{all possible } x_i} P(x_i) = 0.125 + 0.375 + 0.375 + 0.125 = 1.0$

Example 2: A blue tetrahedral die has the numbers 1, 3, 5, and 7 on its four faces. A red tetrahedral die has the numbers 2, 4, 6 and 8 on its four faces. The two dies are tossed and the numbers turned face up on the dies are recorded. If T denotes the sum of the two numbers, determine the probability distribution of all possible and determine the expectation value of T, T assuming that the two dies are fair.

Solution

The possible outcomes can be represented in the table below:

Blue\Red	2	4	6	8
1	3	5	7	9
3	5	7	9	11
5	7	9	11	13
7	9	11	13	15

The four faces of a fair die have equal probabilities of turning face up. Therefore, probability of each face turning face up = ¼ =0.25.

T	3	5	7	9	11	13	15	Σ
Frequency, $f(T)$	1	2	3	4	3	2	1	16
Probability, $P(T) = \dfrac{f(T_i)}{\sum f(T_i)}$	1/16	2/16	3/16	4/16	3/16	2/16	1/16	1.0
$T*P(T)$	3/16	10/16	21/16	36/16	33/16	26/16	15/16	9.5

Expectation value of $T = \displaystyle\sum_i T_i P_i(T) = \frac{1}{16}\left(3+10+21+36+33+26+15\right) = \frac{144}{16} = 9.5$

Example 4: A discrete random variable *x* has a probability distribution given in the table below.

x	-2	-1	1	2	5	10	Σ
P(x)	0.1	0.2	0.3	0.2	0.1	0.1	1.0

Determine the expectation value of $100/x^2$.

Solution

x	-2	-1	1	2	5	10	Σ
P(x)	0.1	0.2	0.3	0.2	0.1	0.1	1.0
$y(x) = 100/x^2$	25	100	100	25	4	1	255
$y(x)*P(x)$	2.5	20	30	5	0.4	0.1	58

$$E[y(x)] = \sum_i y_i(x)P_i(x) = \sum_i \frac{100}{x_i^2} P_i(x) = \frac{100}{-2^2}0.1 + \frac{100}{-1^2}0.2 + \frac{100}{1^2}0.3 + \frac{100}{2^2}0.2 + \frac{100}{5^2}0.1 + \frac{100}{10^2}0.1$$
$$E[y(x)] = \sum_i y_i(x)P_i(x) = 2.5 + 20 + 30 + 5 + 0.4 + 0.1 = 58$$

Example 5: A health department record shows that patients suffering from a certain disease has a probability of 75% dying from it. What is the probability that four of six randomly selected patients will recover?

Solution

For 6 randomly selected patients with four persons recovering, number of patient dying = 2.
The probability is given by Eqn (6), $P(x) = {}^nC_x p^x (1-p)^{(n-x)}$ or

$$P(2) = {}^6C_2 0.75^2 (1-0.75)^4 = \frac{6!}{2!4!} 0.75^2 0.25^4 = 0.0330$$

In *Excel*, there is a ***binomial distribution*** function called BINOMDIST(*x*, *n*, *p*, *cumulative*) which can be used to calculate binomial probabilities. In the string specifications, *x*, *n*, and *p* correspond to the number of successes, number of trials, and probability of success, respectively. Permissible entry for *cumulative* is either *true* or *false*. If *true* is specified, BINOMDIST will return the *cumulative* distribution function, which is the probability that there are at most *x* successes. If *false* is specified, BINOMDIST will return the probability function, which is the probability that there are *x* successes. In Example 5 above, the use of BINOMDIST(2, 6, 0.75, *false*) gives a value of 0.032959, in exact agreement with the hand calculation.

(B) Poisson Distribution Function

Poisson distribution is a discrete probability distribution that refers to the number of events (or successes) within a specific time period or region of space. For example: (i) the number of cars arriving at a service station in 1 hour (the time interval is 1 hour), (ii) the number of defective units in a box of 12 product units (the number interval is 12), and the number of accidents in 1 day on a particular stretch of highway (the interval is defined by both time, 1 day, and space, the particular stretch of highway). The Poisson distribution is given by $P(x) = \dfrac{e^{-\mu}\mu^{x}}{x!}$, where μ is the mean number of successes within the interval, and x is the specified number of successes.

Example 6: A company makes electric motors. The probability an electric motor is defective is 0.01. What is the probability that a sample of 300 electric motors will contain exactly 5 defective motors?

Solution

The average number of defectives in 300 motors is $\mu = 0.01 \times 300 = 3$

The probability of getting 5 defectives is $P(x=5) = \dfrac{e^{-3}3^{5}}{5!} = 0.10082$

If we solve the problem using binomial distribution, we would obtain with $n = 300$, $x = 5$, $p = 0.01$ and $q = 1 - p = 0.99$, $P(x = 5) = {}^{300}C_5(0.01)^5(0.99)^{295} = 0.10099$. Both results are acceptable.